

Run Report: 2026090511055203bd3e

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1. Configuration

parameter	value
numrobots	13
seed	42
stopmode	workload
targettasks	2000
maxticks	100000
scheduler	auction
pathplanner	bfs
conflictmanager	localyield
demandmode	legacy
layoutpreset	default
chargercapacity	4
failureenabled	False
fastmode	True

2. Performance KPIs

metric	value
stopreason	targetreached
simulationticks	5292
taskscreated	2010
tasksassigned	2214
taskscompleted	2000
tasksfailed	0
averagethroughput	0.378
averagewaittime	6.185
averagecycletime	30.460
averagerobotutilization	0.840
totalreplans	952
totalblockedticks	4139
blockedtimeseconds	1241.700
chargingevents	244
chargerwaitticks	97
batterytaskinterruptions	204
failuretaskinterruptions	0
taskreassignments	200

3. Distributions

metric	count	mean	median	p90	p95	max
taskwaitingtime	2010	4.146	0.000	10.000	19.000	252.000
taskcycletime	2000	30.460	26.000	56.100	71.000	310.000
robotblockedepisodeticks	13	5206.769	5215.000	5257.000	5261.800	5269.000
systemactivefraction	5292	0.710	0.692	0.846	0.923	1.000
robotutilization	13	0.840	0.843	0.853	0.856	0.860

4. Spatial Bottleneck Analysis

Coverage: 13/13 blocked events (100.0%).

rank	cell	blockedevents	blockedticks
1	(4, 9)	2	10416
2	(6, 4)	1	5269
3	(19, 1)	1	5257
4	(10, 11)	1	5257
5	(7, 4)	1	5242

Blocked ticks are attributed to the cell where blocking started. Hotspots near chargers indicate capacity problems; hotspots at aisle intersections indicate traffic-control problems.